CM0246 Discrete Structures
Mathematical Induction

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Preliminaries

Textbook

Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Motivation

Exercise

Conjecture a formula for the sum of the first n positive odd integers.

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Question

Let $P(n)$ be a propositional function. How can we proof that $P(n)$ is true for all $n \in \mathbb{Z}^+$?

Principle of Mathematical Induction

Proof by mathematical induction

Let $P(n)$ be a propositional function.

To prove that $P(n)$ is true for all $n \in \mathbb{Z}^+$, we must make two proofs:

- Basis step: Prove $P(1)$

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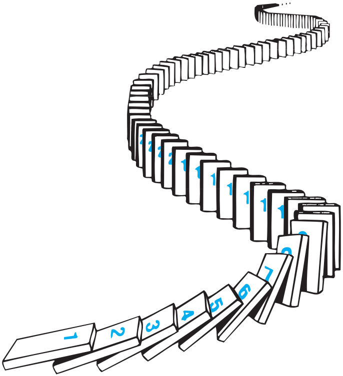
- Inductive step: Prove $P(k) \to P(k+1)$ for all $k \in \mathbb{Z}^+$

  $P(k)$ is called the inductive hypothesis.

Principle of Mathematical Induction

How mathematical induction works†



†Figure source: (Rosen [1988] 2012, § 5.1, Fig. 2).

Principle of Mathematical Induction

Definition (principle of mathematical induction)

Inference rule version:

$$\dfrac{P(1) \qquad \begin{array}{c}[P(k)]\\ \vdots \\ P(k+1)\end{array}}{P(n)}\ \text{(PMI)}$$

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Axiom (or theorem) version: Let P be a propositional function (predicate). Then

$$[P(1) \wedge \forall k(P(k) \to P(k+1)] \to \forall n P(n) \quad \text{(PMI)}$$

Principle of Mathematical Induction

Methodology for proving by mathematical induction

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3. Prove the induction step, i.e. $\forall k(P(k) \to P(k+1))$.

   Remark: In this proof you need to use the inductive hypothesis $P(n)$.

Principle of Mathematical Induction

Methodology for proving by mathematical induction

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2. Prove the basis step, i.e. $P(1)$.
3. Prove the induction step, i.e. $\forall k(P(k) \rightarrow P(k+1))$.

   Remark: In this proof you need to use the inductive hypothesis $P(n)$.
4. Conclude $\forall n P(n)$ by the principle of mathematical induction.

Principle of Mathematical Induction

Example

Prove that the sum of the first n odd positive integers is n^2.[†]

Whiteboard.

[†] Historical remark. From 1575, it could be the first property proved using the PMI (Gunderson 2011, § 1.8).

Principle of Mathematical Induction

Example

Prove that if $n \in \mathbb{Z}^+$, then

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Principle of Mathematical Induction

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Proof

1. $P(n)$: $1 + 2 + 3 + \cdots + n = \dfrac{n(n+1)}{2}$.

Principle of Mathematical Induction

Example

Prove that if $n \in \mathbb{Z}^+$, then

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof

1. $P(n)$: $1 + 2 + 3 + \cdots + n = \dfrac{n(n+1)}{2}$.
2. Basis step $P(1)$: $1 = \dfrac{1(1+1)}{2}$.

Continued on next slide

Principle of Mathematical Induction

Proof (continuation)

3. Inductive step:

   Inductive hypothesis $P(k)$: $1 + 2 + 3 + \cdots + k = \dfrac{k(k+1)}{2}$.

   Let's prove $P(k+1)$:

$$\begin{aligned} 1 + 2 + 3 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) && \text{(by IH)} \\ &= (k+1)\left(\frac{k}{2} + 1\right) && \text{(by arithmetic)} \\ &= \frac{(k+1)(k+2)}{2} && \text{(by arithmetic)} \end{aligned}$$

Principle of Mathematical Induction

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&= \frac{(k+1)(k+2)}{2} && \text{(by arithmetic)}
\end{aligned}
$$

4. $\forall n P(n)$ by the principle of induction mathematical. ■

Principle of Mathematical Induction

Example

Prove that if $n \in \mathbb{N}$, then

$$2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$$

Proved on next slide

Principle of Mathematical Induction

Proof

1. $P(n)$: $2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$

Principle of Mathematical Induction

Proof

1. $P(n)$: $2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$
2. Basis step $P(0)$: $2^0 = 1 = 2^{0+1} - 1$.

Principle of Mathematical Induction

Proof

1. $P(n)$: $2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$
2. Basis step $P(0)$: $2^0 = 1 = 2^{0+1} - 1$.
3. Inductive step:
   Inductive hypothesis $P(k)$: $2^0 + 2^1 + 2^2 + \cdots + 2^k = 2^{k+1} - 1$
   Let's prove $P(k+1)$:

$$\begin{aligned} 2^0 + 2^1 + 2^2 + \cdots + 2^k + 2^{k+1} &= 2^{k+1} - 1 + 2^{k+1} && \text{(by IH)} \\ &= 2(2^{k+1}) - 1 && \text{(by arithmetic)} \\ &= 2^{k+2} - 1 && \text{(by arithmetic)} \end{aligned}$$

Principle of Mathematical Induction

Proof

1. $P(n)$: $2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$
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   Let's prove $P(k+1)$:

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4. $\forall n P(n)$ by the principle of induction mathematical. ■

Strong Induction

Proof by strong (or course-of-values) induction

Let $P(n)$ be a propositional function.

To prove that $P(n)$ is true for all $n \in \mathbb{Z}^+$, we must make two proofs:

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Strong Induction

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Let $P(n)$ be a propositional function.

To prove that $P(n)$ is true for all $n \in \mathbb{Z}^+$, we must make two proofs:

- Basis step: Prove $P(1)$
- Inductive step: Prove $[P(1) \wedge P(2) \wedge \cdots \wedge P(k)] \rightarrow P(k+1)$ for all $k \in \mathbb{Z}^+$

  The (strong) inductive hypothesis is given by

  $$P(j) \text{ is true for } j = 1, 2, \ldots, k.$$

Strong Induction

Definition ([strong induction)

Inference rule version:

$$\frac{P(1) \qquad \forall k[(P(1) \wedge P(2) \wedge \cdots \wedge P(k)) \rightarrow P(k+1)]}{\forall n P(n)} \text{ (strong induction)}$$

Strong Induction

Example (a part of the fundamental theorem of arithmetic)

Prove that if n is an integer greater than 1, either is prime itself or is the product of prime numbers.

Proved on next slide

Strong Induction

Proof

1. $P(n)$: n is prime itself or it is the product of prime numbers.

Strong Induction

Proof

1. $P(n)$: n is prime itself or it is the product of prime numbers.
2. Basis step $P(2)$: 2 is a prime number.

Strong Induction

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   Inductive hypothesis: $P(j)$ is true for $j = 1, 2, \ldots, k$.

Strong Induction

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1. $P(n)$: n is prime itself or it is the product of prime numbers.
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   Inductive hypothesis: $P(j)$ is true for $j = 1, 2, \ldots, k$.
   Let's prove that $k + 1$ satisfies the property:

Strong Induction

Proof

1. $P(n)$: n is prime itself or it is the product of prime numbers.
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3. Inductive step:
   Inductive hypothesis: $P(j)$ is true for $j = 1, 2, \ldots, k$.
   Let's prove that $k + 1$ satisfies the property:
   3.1 If $k + 1$ is a prime number then it satisfies the property.

Strong Induction

Proof

1. $P(n)$: n is prime itself or it is the product of prime numbers.
2. Basis step $P(2)$: 2 is a prime number.
3. Inductive step:
   Inductive hypothesis: $P(j)$ is true for $j = 1, 2, \ldots, k$.
   Let's prove that $k + 1$ satisfies the property:
   3.1 If $k + 1$ is a prime number then it satisfies the property.
   3.2 If $k + 1$ is a composite number:
   $k + 1 = ab$ where $2 \leq a \leq b < k + 1$. Since $P(a)$ and $P(b)$ by the inductive hypothesis, then $P(k + 1)$.

Strong Induction

Proof

1. $P(n)$: n is prime itself or it is the product of prime numbers.
2. Basis step $P(2)$: 2 is a prime number.
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   Inductive hypothesis: $P(j)$ is true for $j = 1, 2, \ldots, k$.
   Let's prove that $k + 1$ satisfies the property:
   3.1 If $k + 1$ is a prime number then it satisfies the property.
   3.2 If $k + 1$ is a composite number:
   $k + 1 = ab$ where $2 \leq a \leq b < k + 1$. Since $P(a)$ and $P(b)$ by the inductive hypothesis, then $P(k + 1)$.
4. $P(n)$ is true for all integer n greater than 1 by strong induction. ■

First-Order Peano Arithmetic

Giuseppe Peano
(1858 – 1932)

Axioms of first-order Peano arithmetic[†]

$$\forall n.\ 0 \neq n'$$
$$\forall m \forall n.\ m' = n' \rightarrow m = n$$
$$\forall n.\ 0 + n = n$$
$$\forall m \forall n.\ m' + n = (m + n)'$$
$$\forall n.\ 0 * n = 0$$
$$\forall m \forall n.\ m' * n = n + (m * n)$$

[†] See, for example, (Hájek and Pudlák [1993] 1998).

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For all formulae A,

$$[A(0) \wedge (\forall n.\ A(n) \rightarrow A(n'))] \rightarrow \forall n A(n)$$

[†] See, for example, (Hájek and Pudlák [1993] 1998).

First-Order Peano Arithmetic

Theorem

The principle of mathematical induction and strong induction are equivalent.[†]

[†] See, for example, (Gunderson 2011).

References

David S. Gunderson (2011). Handbook of Mathematical Induction. Chapman & Hall (cit. on pp. 15, 40).

Petr Hájek and Pavel Pudlák [1993] (1998). Metamathematics of First-Order Arithmetic. Second printing. Springer (cit. on pp. 38, 39).

Kenneth H. Rosen [1988] (2004). Matemática Discreta y sus Aplicaciones. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).

— [1988] (2012). Discrete Mathematics and Its Applications. 7th ed. McGraw-Hill (cit. on p. 7).